

SOES 3042/6025 – Assignment 1

This assignment accounts for **40%** of your total module grade. The deadline to submit your answer is **Monday 2 pm 10th November 2026**. Submit your answer as one PDF file via Turnitin on the module's site and name the file: "**SOES3042-A1_StudentID_LastName.pdf**". Replace "StudentID" with your 8-digit student ID and "LastName" with your last name. If you are taking SOES6025, replace "3042" with "6025".

Complete this assignment by following the instructions below

(1) Task: Critical Review of an AI-Generated Report. Instead of writing a standard report from scratch, you are provided with a "Draft Short Report" generated by a GenAI model. Your task is to grade this report (out of 100) and provide a detailed critique.

- **The Deliverable:** A written assessment that identifies specific errors, inconsistencies, and hallucinations in the AI's answer.
- **The Approach:** Do not simply correct grammar or formatting. You must look for technical flaws (e.g., mathematical derivation, statistical assumptions, and visual anomalies in figures).
- **Evaluation Criteria:** High marks will be awarded to those who can spot subtle but fatal errors, explain why they are wrong based on course theory, and propose a valid fix.

(2) Formatting & Length: Using the provided Word template, your critique for each problem must not exceed one side of an A4 page.

- **Structure:** Start with your assigned score and then dive directly into bullet points listing major errors you found and why. For example:
 - *The report claims to have removed seasonal cycles, but seasonality retains in EOF PCs.*
- **DO NOT** write any generic summaries, which won't affect your marks positively and will very likely take up your limited space.
- **Evidence:** Please include your own corrected figures or calculations to contrast with the AI's errors. These corrections count towards the page limit, so be concise. Use single line spacing and a font size no smaller than 11 pt. You may adjust the size of the answer boxes for the "short report" and "figures" for each problem's evaluation.

(3) Verification & Code

- **Recommendation:** While you are not explicitly required to submit a full code report, it is strongly recommended that you perform the analysis yourself to establish the "Ground Truth." Without generating your own results, it will be extremely difficult to detect the hidden errors in the AI report.
- **Submission:** If you used code to verify the results and generate corrected figures, please submit your Jupyter Notebook as a supplementary file ("SOES3042-A1_StudentID_LastName.ipynb"). It won't be graded but this serves as evidence of your verification process.

ARC Policy for SOES 3042/6025 Assignment 1

Collaboration Policy

You may discuss the underlying theoretical concepts with your classmates (e.g., the mathematical properties of EOFs). However, the identification of specific errors within the provided AI report and the writing of your critique must be done strictly independently. In other words, **sharing the specific flaws you have found with others is strictly prohibited.**

- **Allowed:** Discussing *"How should the explained variance be calculated theoretically?"*
- **Prohibited:** Telling a classmate *"The report is wrong because it didn't ..."*

Disclosure Requirements: You **MUST** disclose the names of all collaborators. For this assignment, a collaborator is any individual (including classmates, tutors, or external persons) with whom you have discussed the assignment's in any form.

GenAI usage Policy

Activities	Tier	Boundary
Coding	2	Restricted Use Allowed: You may use GenAI to explain error messages, debug specific syntax errors, or look up library documentation (e.g., "Why is my np.linalg.svd returning this shape?"). Prohibited: You can NOT ask GenAI to generate the full analysis script or write logical blocks of code for you (e.g., "Write a script to perform EOF analysis on this dataset" is forbidden).
Critique & Evaluation	1	Strictly Prohibited You must NOT paste the provided AI report (or parts of it) into any GenAI tool (e.g., ChatGPT, Claude, Gemini) to ask for a critique, summary, or error detection. The identification of errors must rely solely on your own judgment and verification.
Writing & Drafting	1	Strictly Prohibited You must NOT use GenAI to draft, write, rewrite, paraphrase, or expand your sentences. The critique must be written entirely in your own words to ensure technical precision. Since GenAI tools excel at style but often fail at substance, we will not be grading on flow or vocabulary. We are grading strictly on your ability to identify physical and mathematical fallacies.

Disclosure Requirements: You **MUST** complete, sign, and submit an ARC declaration at the last page of SOES3042-A1_StudentID_LastName.pdf to confirm your understanding and agreement with these policies. The form can be found at the end of the Word answer template. Failure to submit this declaration will result in your assignment not being graded.

⚠ **A Note on "AI-on-AI" Critique:** GenAI tools are probabilistic, not analytical. If you ask a GenAI tool to review the provided draft report, it is statistically likely to "hallucinate" that the report is correct because the text sounds plausible. **Using GenAI to critique the report is not only a breach of Tier 1 policy but is also the fastest way to fail this assignment.** We have designed specific errors that are "invisible" to current LLMs but obvious to a human who understands the physics.

Be advised: any detected breach of these policies, including undisclosed collaboration or the use of Generative AI, will be treated as academic misconduct under the ARC framework. Such breaches will result in severe penalties, including but not limited to receiving a grade of zero for this assignment and potential further disciplinary action.

The assignment

In our final lecture, we used the EOF technique to uncover significant patterns of sea-level pressure variability in the Tropical Pacific — a key atmospheric variable. This assignment delves into the oceanic component of climate variability by analysing sea-surface temperature (SST) datasets. Through this exercise, you will replicate the methodologies employed by scientists to identify and analyse El Niño and La Niña events. These phenomena, along with their atmospheric counterparts, constitute the El Niño-Southern Oscillation (ENSO), one of the most significant modes of variability influencing global weather and climate.

You will apply the skills and knowledge acquired throughout the course to complete this assignment. The dataset you will need to use are listed in below:

filename	location	Type of data
ERSST5_1979_to_2023.nc	data/	Gridded monthly sea surface temperature at 2°x2° resolution
SLP_Darwin/ Tahiti.txt	data/	Monthly sea-level pressure at the Darwin/Tahiti station

Problem 1. Nino3 Index and Southern Oscillation

The term "El Niño" (Spanish for "the boy" or "the Christ child") was first used by Peruvian fishermen in the late 19th century to describe the appearance of unusually warm water in the eastern Pacific Ocean around Christmas time. The Nino3 index, calculated by averaging SST anomalies over the region bounded by [5°N-5°S and 210°-270°E](#), is a widely used measure of the oceanic component of ENSO.

(a) Remove seasonal cycles from the loaded monthly SST and calculate the Nino3 index for each month. Create a time series plot of the Nino3 index.

(b) Plot a histogram of the monthly Nino3 index values. Visually, does the distribution appear to be normal? Can you then find a quantitative approach that we have discussed in our module to test whether the monthly Nino3 index follows a normal distribution?

(c) The atmosphere and the ocean are closely coupled. Please calculate the Southern Oscillation (SO) index by subtracting the sea-level pressure (SLP) anomalies at Darwin from the SLP at Tahiti. Plot it on top of the Nino3 index calculated in question (a).

(d) Plot the two indices as a scatter plot. Are the two variables any how correlated? Can you find a quantity to quantify the correlation? Is it statistically significant? Consider which approach to use. What assumptions are involved? Are they valid?

Problem 2. EOF Analysis of Tropical SST anomalies

Building on our understanding of Empirical Orthogonal Functions (EOF) from Lecture 9, this problem focuses on analysing detrended sea-surface temperature (SST) anomalies in the tropics (0-360°E, 20°S-20°N). EOF analysis will enable us to identify dominant patterns of variability within the dataset and understand their temporal evolution. Please:

(a) Remove the seasonal cycle and long-term warming trends of SST at each location and then perform an EOF analysis on the SST anomalies. Should each grid point have equal weight in an EOF analysis? Why?

(b) After you perform the EOF analysis, plot the time series and associated spatial patterns of the first two EOF modes. Compare with the Nino3 index calculated in Problem 1, which mode corresponds to El Niño events? How much variance is explained by these two modes?

(c) Create a scatter plot of PC2 (y-axis) versus PC1 (x-axis) during the December to February timeframe. Do you expect any significant relationship to emerge between the two PCs during winter? Perform a quadratic regression of PC2 against PC1 using ordinary least squares (OLS) and plot the results. Is the fitted model overall statistically significant? What is the assumption in this fitting? Does it hold? How would it affect the interpretation of significance?

(d) Also explore a linear and cubic fit for the relationship between wintertime PC1 and PC2, selecting the best model among linear, quadratic, and cubic using a quantitative approach. (SOES6025 only, if you are taking SOES3042, your critique doesn't need to include this question d).

AI Generated Report

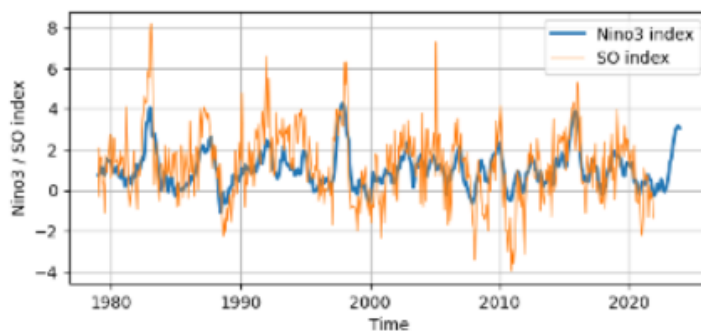
Short Report for Problem 1

(a, c) The seasonal cycle is removed by subtracting the 1979-2023 annual climatology from each month's SST and SLP data. Algorithmic mean of SSTs over 5°N-5°S, 210°-270°E is then calculated as the Nino 3 index. Both indices are aligned and then plotted on top of each other (Fig.1a).

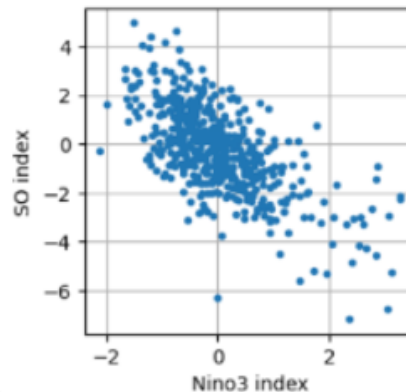
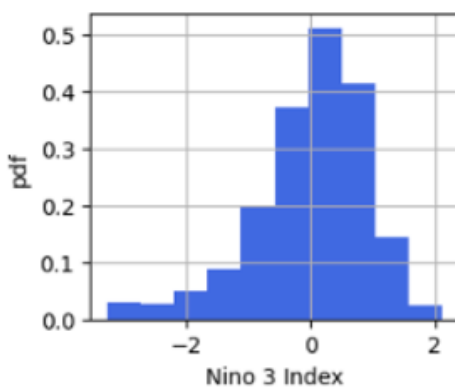
(b) The histogram of Nino 3 index (Fig. 1b) shows negative skewness of -0.97, indicating this index may not follow a Gaussian distribution. Normality can be further assessed using a Lilliefors test, which compares the minimum distance between the data-based empirical PDF and that of a standard Gaussian distribution. To perform this test, I call the 'lilliefors' function with the blue time series shown in Figure 1 as the input, which returns a p-value of 0.001. A p-value smaller than 0.05 automatically means that the null hypothesis is rejected and that the data do not follow a Gaussian distribution.

(d) Nino 3 and SO indices are plotted against each other (Fig. 1c). Because no apparent non-linear relationship is observed between the two indices, we can very safely use Pearson's correlation coefficient. To get a robust estimate, I standardised both indices before calling the function, which yields an r value of 0.66 and a p-value of 0.0, suggesting that the two indices are highly positively correlated. However, the test in correlation function assumes that both datasets follow Gaussian distributions, which should hold for both SST and SLP because these are monthly averaged values, and we have learned the central limit theorem, which says that averaged quantities (with ~30 samples) will approach normal distributions. As a result, we should be able to trust the python package output without much concern.

Figures (with captions) for Problem 1



Figures 1. (a) Monthly Nino 3 and SO index. (b) Histogram of the Nino 3 index. (c) Scatter plot of the two indices against one another.



Short Report for Problem 2

(a) Long-term linear trends are removed from monthly SST anomalies, which are then reshaped into a 3420 by 540 matrix, and an EOF is performed using the 'syd' function. Because the area of a grid box increases towards the pole, data should be scaled by the cosine latitude such that each grid box contributes equally in the analysis.

(b) The first two EOF modes should correspond to the first two rows of the U matrix returned from the SVD decomposition. I plot the standardised rows of the V matrix as the principal component (Fig.2a), and regress SSTs at each location to obtain the spatial patterns of individual modes (Fig.2b). The second mode has a Pearson's correlation of 0.97 with the Nino 3 index calculated in the last problem, suggesting that the second mode should be El-Nino. The explained variance of the j^{th} mode is determined by the diagonal elements in matrix D, using $d_j / \sum_{i=1 \rightarrow N} d_i$. The cumulative explained variance is shown in Fig.2c, with the first three modes explaining a total of 50% variance.

(c) We do not expect the PCs generated by EOF to be anyhow correlated, a key characteristic from EOF's orthogonal feature. Fitting to the wintertime data confirms this statement, that Pearson's correlation between wintertime PC1 and PC2 is strictly zero. Moving beyond fitting a line, the result of a quadratic fitting using OLS is shown in Fig.2d, this model fit is significant because the p-value of x1, as returned by the statsmodels summary, is nearly zero. A key assumption behind this fitting is that data follow Gaussian distributions, which we know does not hold from the last problem. This could affect the reliability of the p-value; yet given the fact that p-value is already very small, we should be able to confidently trust the python package that the fitting is statistically highly significant.

(d) Among the linear, quadratic, and cubic models, the quadratic model does the best job in depicting the relationship between wintertime PC1 and PC2, because the RMSE of a linear fitting is 0.04, decreases to 0.03 in quadratic and stop decreasing further for the cubic model.

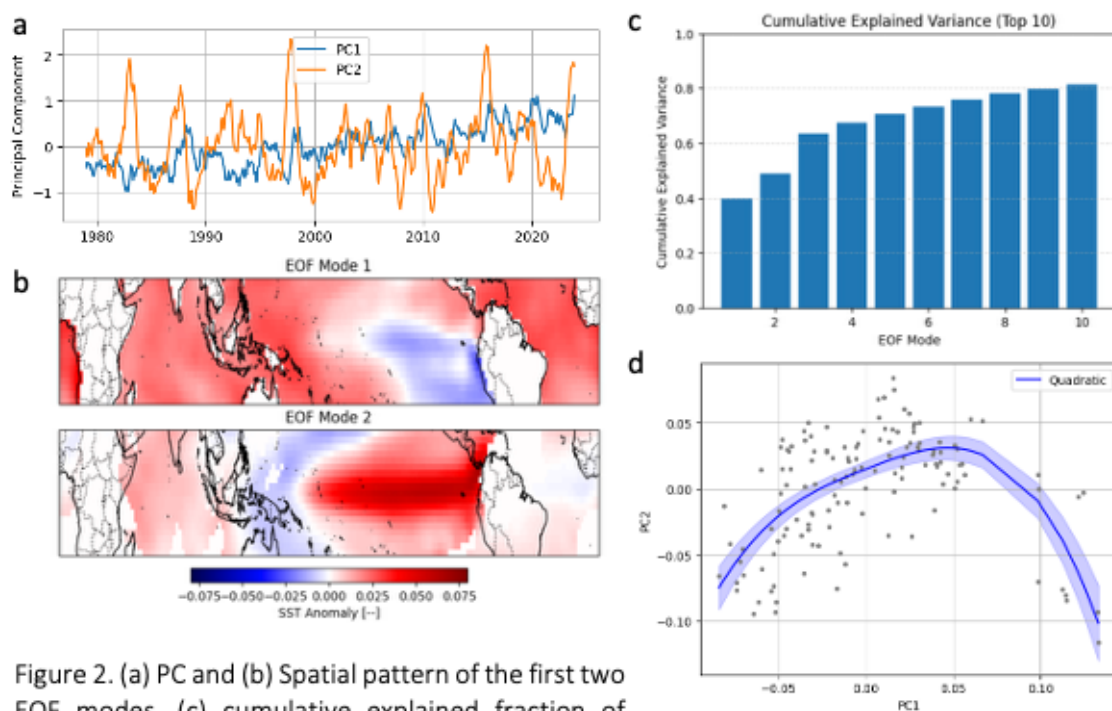
Figures (with captions) for Problem 2

Figure 2. (a) PC and (b) Spatial pattern of the first two EOF modes. (c) cumulative explained fraction of variance. (d) PC1 vs PC2, together with quadratic fitting (blue) and 95 c.i. indicated by shading.